LEVEL PENDING CLASSIFICATION

DETECTIONS DISABLED

SQL Server DR PoC Requirements

This document was submitted to our Cohesity account manager on 20/05/2026

North Star: For the production-tier SQL Server estate currently protected by Veritas NetBackup, achieve a recoverable state of ≤2-hour RTO and ≤15-minute RPO under an infrastructure-failure / site-loss event, evidenced by a technical PoC.

Decision Frame: Decide whether to adopt Cohesity DataProtect (self-hosted in AWS) as the replacement for Veritas NetBackup for the in-scope production SQL Server estate.

Recovery Scenario Framework

Dimension	Logical Recovery	Disaster Recovery	Data Provisioning
Context	Human Error / Bug	Catastrophic Event (AZ loss; ransomware)	Planned Request (dev refresh / legal hold / audit)
Trigger	Accidental DML or application bug corrupting live data	Infrastructure failure or cyber attack	Planned request from development, legal, or audit team
System State	System UP; data INVALID; Cohesity cluster healthy	System DOWN; source AZ unavailable; primary Cohesity cluster may be gone	System variable; production running; need static isolated copy
Goal	Recover to the last valid point; preserve all valid work after that point	Restore SQL service with full identity; meet RTO and RPO	Create an isolated, consistent copy of production data for non-production use
Urgency	Variable (tends to be high)	Critical (≤2-hour RTO)	Planned (no strict urgency)

Mechanism	Cohesity T-Log PITR restore	Cohesity identity-preserved EC2 + SQL restore to alternate AZ; DataLock for cyber recovery	Cohesity database restore to alternate non-production host
Recovery Scope	Database(s)	EC2 instance + Databases	Database(s)
Restore Target	Same host or alternate host (same AZ)	Different AZ	Non-production EC2 instance (any AZ)

Specification by Example

Feature: SQL Server Recovery with Cohesity DataProtect

```
Background:
  Given our RTO target is ≤2 hours
  And our RPO target is ≤15 minutes (Cohesity VDI protection owns the T-Log chain)
  And Cohesity DataProtect is self-hosted on AWS EC2/EBS
  And all in-scope SQL databases are in FULL recovery model
  And Cohesity SQL VDI protection is the exclusive owner of the transaction-log backup chain
  And Cohesity has network connectivity to all restore targets
```

These scenarios are prerequisites for all logical recovery and disaster recovery scenarios. A failure here stops all further testing.

Rule: Backup reliability must be proven before any restore scenario is tested

This is the prerequisite gate. If Cohesity cannot reliably complete and report backups, no restore scenario can be trusted and the PoC fails here.

```
Scenario: Backup reliability — 48-hour observation window (standalone and AG)
  Given a standalone SQL instance and an Always On AG are registered in Cohesity
  And business-representative workload is applied throughout the window
  And AG AUTOMATED_BACKUP_PREFERENCE and BACKUP_PRIORITY are explicitly configured
  When Cohesity protection runs for a 48-hour window
  Then every scheduled full, differential/incremental, and log backup completes successfully # Completeness
  And no in-scope database is silently skipped # No silent skips
  And every Cohesity-reported backup reconciles with msdb..backupset # Reconciliation
```

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  And RESTORE VERIFYONLY succeeds for every backup file sampled (at least one per database/type) # Verification
  And the T-Log chain is intact (no gaps) for every in-scope database at window close # Chain integrity
  And AG backup honours the configured backup preference and replica priority # AG awareness
```

A backup that reports success but fails on restore is worse than one that fails openly — it creates false confidence. This scenario validates that every Cohesity-reported successful recovery point is genuinely restorable — not just reported as complete.

```
Scenario: Reported-successful backup is restorable — 3/3 runs
  Given the 48-hour backup reliability scenario has passed
  And Cohesity shows successful recovery points (full, diff/incremental, log) for in-scope databases
  And a clean restore-target SQL instance is available on a different host
  When a set of recovery points is selected and each is restored to the clean target
    including at least one full recovery point, one mid-chain point, and one latest-log point
  Then every Cohesity-reported successful recovery point restores without error # Restore success
  And DBCC CHECKDB returns 0 allocation errors and 0 consistency errors for every restored database # Integrity
  And application smoke tests against the restored databases pass # Application
```

Rule: Logical recovery must support point-in-time restore

This proves the RPO claim. If we cannot recover data to within 15 minutes of a failure using only the Cohesity log chain, the ≤15-minute RPO target is not met.

```
Scenario: Point-in-time recovery within 15 minutes of failure — 3/3 runs
  Given the backup reliability scenarios have passed
  And Cohesity is taking log backups at ≤15-minute cadence
  And a known data marker is written to the database within one log-backup interval before T0
  When simulated failure T0 is declared
  And the database is restored to the latest valid point before T0 using the Cohesity chain
  Then the latest recoverable point is no older than 15 minutes before T0 # RPO
  And the data marker is present in the restored database # Data recency
  And DBCC CHECKDB returns 0 allocation errors and 0 consistency errors # Integrity
```

A silent chain break is the most dangerous RPO failure: backups continue to report success while point-in-time recovery becomes impossible. This scenario validates that a chain break is detected and surfaced, never silent.

```
Scenario: T-Log chain break is detected and does not silently corrupt RPO
  Given a database's recovery model is switched from FULL to SIMPLE and back
  When Cohesity attempts to continue the T-Log backup chain
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```
  Then Cohesity detects the chain break
  And either triggers an automatic full backup or raises an observable alert to the operator
  And the chain break is never silent
```

Rule: Disaster recovery must restore the SQL service with identity preserved

This is the RTO proof. The restored SQL service must appear to the domain as the same machine returning online — not a new machine with the right name. If identity is broken, recovery becomes a rebuild.

```
Scenario: Cross-AZ restore with identity preserved within 2 hours — 3/3 runs
  Given the backup reliability scenarios have passed
  And the identity surface (OS, network, SQL instance, SQL security, SQL Agent,
    SQL configuration, system databases, application) is written down and agreed
    before the first run begins
  And a clean target environment exists in a different Availability Zone
  And recovery is designed to proceed from the surviving Cohesity copy
    (the primary Cohesity cluster is assumed unavailable)
  When a simulated AZ-loss event is declared at T0
  And the runbook is executed
  Then a replacement SQL service is reachable in the target AZ # Reachability
  And every layer of the agreed identity surface validates # Identity
  And DBCC CHECKDB returns 0 allocation errors and 0 consistency errors for every restored user database # Integrity
  And the application smoke test passes # Application
  And wall-clock RTO (T0 to smoke-test pass) is ≤2 hours # RTO
```

DataLock is the cyber recovery claim. This scenario validates that immutable recovery points remain intact and restorable even when tested against privileged roles.

```
Scenario: DataLock-protected recovery point survives destructive-role attempt (1/1)
  Given the backup reliability scenarios have passed
  And Cohesity DataLock / WORM immutability is configured for in-scope protection jobs
  And at least one recovery point is protected by DataLock
  And a threat model defines the tested roles
    (backup operator, Cohesity admin etc.)
  When each tested role attempts to delete, shorten the retention of,
    or alter the immutability policy of the protected snapshot
  Then all destructive attempts are blocked # Blocked
  And all attempts are logged in the Cohesity audit log and in CloudTrail # Audit trail
  And the immutable recovery point remains intact and restorable # Intact
  And a restore from the immutable point passes DBCC CHECKDB 0/0 and smoke tests # Restorable
  And the backup-operator role cannot modify DataLock settings # Least privilege
```

Rule: Data provisioning is out of scope for this PoC

Cohesity supports database restore to alternate non-production hosts for dev/test refresh, legal hold, and audit. This relies on the same restore mechanism validated by the backup reliability scenarios. Data provisioning scenarios are not tested in this PoC and are candidates for operational runbooks after the Go/No-Go decision.

Requirements Specification

ID	Requirement
REQ-001	Cohesity SQL protection SHALL complete full, differential/incremental, and T-Log backups with ≥99% success rate and zero silent skips across a 48-hour observation window
REQ-002	Every Cohesity-reported successful backup SHALL restore to a clean host and pass RESTORE VERIFYONLY and DBCC CHECKDB 0/0
REQ-003	Cohesity T-Log PITR SHALL recover any in-scope database to a point no older than 15 minutes before simulated failure, with zero silent chain breaks
REQ-004	Cohesity SHALL restore a SQL service — including full OS, network, SQL, and application identity — to a different Availability Zone within 2 hours, without requiring the primary Cohesity cluster
REQ-005	Cohesity DataLock SHALL prevent deletion or modification of a

	protected recovery point by any tested role; blocked attempts SHALL be logged; the protected point SHALL remain restorable

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Acceptance Criteria

REQ-001: Backup Reliability

- [ ] ≥99% of protection job runs complete successfully in the 48-hour window
- [ ] Zero in-scope databases are silently skipped
- [ ] T-Log chain intact (no unexplained gaps) at window close for every in-scope database
- [ ] RESTORE VERIFYONLY succeeds for every backup file sampled (at least one per database/type)
- [ ] AG backup honours configured backup preference and replica priority
- [ ] Cohesity job export reconciles with msdb..backupset for all in-scope databases

REQ-002: Restore Reliability

- [ ] 3/3 runs: every selected Cohesity-reported successful recovery point restores without error
- [ ] RESTORE VERIFYONLY succeeds on every backup file used in each run
- [ ] DBCC CHECKDB returns 0 allocation errors and 0 consistency errors for every restored database in each run
- [ ] Application smoke tests pass against restored databases in each run

REQ-003: Point-in-Time Recovery (Logical Recovery)

- [ ] 3/3 runs: every in-scope database is recoverable to a point no older than 15 minutes before T0
- [ ] Data marker written within one log-backup interval of T0 is present in the restored database in each run
- [ ] DBCC CHECKDB returns 0/0 on restored databases in each run
- [ ] T-Log chain break test: Cohesity detects a FULL→SIMPLE→FULL chain break and either triggers an automatic full backup or raises an observable alert; a silent chain break is a fail

REQ-004: Cross-AZ Identity-Preserved Restore (Disaster Recovery)

- [ ] 3/3 runs: wall-clock RTO (T0 to smoke-test pass) ≤2 hours
- [ ] OS identity (hostname, domain membership) validates in each run
- [ ] Network identity (IP, FQDN resolution) validates in each run
- [ ] SQL Instance identity (@@SERVERNAME, instance name, port) validates in each run
- [ ] SQL Security (logins, SIDs, service accounts) validates in each run
- [ ] SQL Agent (job definitions, schedules, enabled/disabled state) validates in each run
- [ ] SQL Configuration (linked servers, certificates) validates in each run
- [ ] System databases (master, msdb, model) restored; SQL starts and behaves correctly
- [ ] Application smoke test passes in each run
- [ ] DBCC CHECKDB returns 0/0 on all restored user databases in each run
- [ ] Recovery proceeds without the primary Cohesity cluster

REQ-005: DataLock Immutability (Disaster Recovery — cyber)

- [ ] All destructive attempts from all tested roles are blocked

- [ ] All blocked attempts are logged in Cohesity audit log and CloudTrail
- [ ] Alerts fire (where configured) on blocked destructive attempts
- [ ] Immutable recovery point restores successfully after all destructive attempts
- [ ] Restore from immutable point passes RESTORE VERIFYONLY, DBCC CHECKDB 0/0, and smoke tests
- [ ] Backup-operator role cannot modify DataLock settings or retention policy

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Traceability Matrix

Scenario	Category	REQ-001	REQ-002	REQ-003	REQ-004	REQ-005
48-hour backup reliability	Backup Reliability	✓				
Backup → restore validation (3/3)	Backup Reliability		✓			
PITR ≤15 minutes (3/3)	Logical Recovery			✓		
T-Log chain break detection	Logical Recovery			✓		
Cross-AZ restore ≤2 hours (3/3)	Disaster Recovery				✓	
DataLock immutability (1/1)	Disaster Recovery					✓

Glossary

Term	Definition
Logical Recovery	Recovery from data corruption caused by human error or application bug, where infrastructure

	remains intact. Uses point-in-time restore to recover to the last valid state.
Disaster Recovery	Recovery from catastrophic events (AZ loss or ransomware) that take infrastructure offline. Requires restoring the SQL service — including full identity — to an alternate location.
Data Provisioning	Planned copying of production data to non-production environments for development, testing, legal hold, or audit. Not tested in this PoC.
PITR	Point-in-Time Recovery — restoring a SQL database to a specific point before a failure using the transaction-log backup chain.
RTO	Recovery Time Objective — maximum acceptable time to restore service after disruption. Target: ≤2 hours.
RPO	Recovery Point Objective — maximum acceptable data loss measured in time. Target: ≤15 minutes.
Identity Surface	The set of identity items (OS, network, SQL instance, SQL security, SQL Agent, SQL configuration, system databases, application) that must validate after a disaster recovery restore. Agreed in writing before the first cross-AZ run.
DataLock	Cohesity's WORM / immutability feature. Prevents deletion or modification of a protected recovery point until a defined expiry, including by administrators.
T-Log Chain	The unbroken sequence of transaction-log backups that enables point-in-time recovery. A chain break makes PITR impossible before the break without a new full backup.